CSE 102 : Data Structure and Algorithms
Midsem Lab Exam
Set 3 : MM 10

Instructions

- Students are supposed to attempt only one of the following questions
- **STL cannot be used**
- Make sure that your code compiles. Zero marks will be given if the code dosent compile.
- Name your cpp file as "**<roll No>.cpp**". Please submit only your .cpp files.
- Your final submission should be the fully executable code. You may/may not copy the sample code provided on page 2.
- We will be running all submissions through AI sensitive Plag detection tools.

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1. Implement Stack using a link list

Implement Stack data structure using link list. You should implement the push() and pop() functionalities. Write a driver which demonstrates how these two functions are working in the stack.

Input Format

- A prompt should ask the user to enter if they want to "push" a value on stack, "pop" a value from stack or "exit". Based on the option chosen by the user, the code should proceed until the user chooses "exit".

Output Format

Pushed <value>
Popped <value>

Example

__Input__:
```
Choose one of the following: 1. Push, 2. Pop, 3. Exit:1
20
Choose one of the following: 1. Push, 2. Pop, 3. Exit:1
21
Choose one of the following: 1. Push, 2. Pop, 3. Exit:2
Choose one of the following: 1. Push, 2. Pop, 3. Exit:2
Choose one of the following: 1. Push, 2. Pop, 3. Exit:2
Choose one of the following: 1. Push, 2. Pop, 3. Exit:3
```

__Output__:
```
Pushed 20
Pushed 21
Popped 21
Popped 20
Stack Empty
```

Constraints

- All values entered into the stack will be int.

OR

2. Quick Sort

Given an array of size n, write a code to sort the array in descending order using Quick Sort. Please take the first element as pivot.

Input Format

- The first line of the input contains the integer `n`, the number of elements in the array
- Second line contains `n` space separated integers

Output Format

The middle node of the list.

Example

__Input__:
```
6
1 2 3 4 5 6
```

__Output__:
```
6 5 4 3 2 1
```

Constraints

- `1 <= n <= 100`
- `-100 <= val <= 100`, where `val` is the value of a node.All values are int.